NATIONAL ELIGIBILITY CUM ENTRANCE TEST

NEET (UG), 2015

(CODE:C)

1. A photoelectric surface is illuminated successively by monochromatic light of wavelength λ and $\frac{\lambda}{2}$. If the maximum kinetic energy of the emitted photoelectrons in the second case is 3 times that in the first case, the work function of the surface of the material is :

(h = Planck's constant, c = speed of light)

(1) $\frac{2hc}{\lambda}$

(2) $\frac{hc}{3\lambda}$

(3) $\frac{hc}{2\lambda}$

(4) $\frac{hc}{\lambda}$

2. The input signal given to a CE amplifier having a voltage gain of 150 is $V_i = 2\cos\left(15t + \frac{\pi}{3}\right)$. The corresponding output signal will be :

(1) $2\cos\left(15t + \frac{5\pi}{6}\right)$

(2) $300\cos\left(15t + \frac{4\pi}{3}\right)$

(3) $300\cos\left(15t + \frac{\pi}{3}\right)$

(4) $75\cos\left(15t + \frac{2\pi}{3}\right)$

A series R-C circuit is connected to an alternating voltage source. Consider two situations :

(a) When capacitor is air filled.

(b) When capacitor is mica filled.

Current through resistor is i and voltage across capacitor is V then :

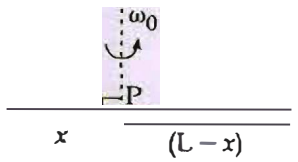
(1) $i_a > i_b$

(2) $V_a = V_b$

(3) $V_a < V_b$

(4) $V_a > V_b$

4. Point masses m_1 and m_2 are placed at the opposite ends of a rigid rod of length L, and negligible mass. The rod is to be set rotating about an axis perpendicular to it. The position of point P on this rod through which the axis should pass so that the work required to set the rod rotating with angular velocity ω_0 is minimum, is given by :



(1) $x = \frac{m_2}{m_1}L$

(2) $x = \frac{m_2 L}{m_1 + m_2}$

(3) $x = \frac{m_1 L}{m_1 + m_2}$

(4) $x = \frac{m_1}{m_2}L$

A parallel plate air capacitor has capacity 'C', distance of separation between plates is 'd' and potential difference 'V' is applied between the plates. Force of attraction between the plates of the parallel plate air capacitor is :

(1) $\frac{CV^2}{d}$

(2) $\frac{C^2V^2}{2d^2}$

(3) $\frac{C^2V^2}{2d}$

(4) $\frac{CV^2}{2d}$

6. An ideal gas is compressed to half its initial volume by means of several processes. Which of the process results in the maximum work done on the gas ?

(1) Isochoric

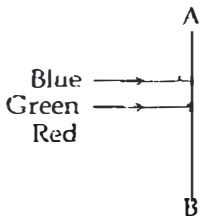
(2) Isothermal

(3) Adiabatic

(4) Isobaric

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7. A beam of light consisting of red, green and blue colours is incident on a right angled prism. The refractive index of the material of the prism for the above red, green and blue wavelength are 1.39, 1.44 and 1.47, respectively.



The prism will :

(1) not separate the three colours at all

(2) separate the red colour part from the green and blue colours

(3) separate the blue colour part from the red and green colours

(4) separate all the three colours from one another

8. Two vessels separately contain two ideal gases A and B at the same temperature, the pressure of A being twice that of B. Under such conditions, the density of A is found to be 1.5 times the density of B. The ratio of molecular weight of A and B is :

(1) 2

(2) $\frac{2}{3}$

(3) $\frac{1}{3}$

(4) 4

A remote - sensing satellite of earth revolves in a circular orbit at a height of 0.25×10^6 m above the surface of earth. If earth's radius is 6.38×10^6 m and $g = 9.8\ ms^{-2}$, then the orbital speed of the satellite is :

(1) 9.13 km s^{-1}

(2) 6.67 km s^{-1}

(3) 7.76 km s^{-1}

(4) 8.56 km s^{-1}

10. The energy of the em waves is of the order of 15 keV. To which part of the spectrum does it belong ?

(1) Ultraviolet rays

(2) γ - rays

(3) X - rays

(4) Infra - red rays

11. A proton and an alpha particle both enter a region of uniform magnetic field B, moving at right angles to the field B. If the radius of circular orbits for both the particles is equal and the kinetic energy acquired by proton is 1 MeV, the energy acquired by the alpha particle will be :

(1) 1.5 MeV

(2) 1 MeV

(3) 4 MeV

(4) 0.5 MeV

12. If vectors $\vec{A} = \cos\omega t\ \hat{i} + \sin\omega t\ \hat{j}$ and $\vec{B} = \cos\frac{\omega t}{2}\ \hat{i} + \sin\frac{\omega t}{2}\ \hat{j}$ are functions of time, then the value of t at which they are orthogonal to each other is :

(1) $t = \frac{\pi}{\omega}$

(2) $t = 0$

(3) $t = \frac{\pi}{4\omega}$

(4) $t = \frac{\pi}{2\omega}$

13. A rectangular coil of length 0.12 m and width 0.1 m having 50 turns of wire is suspended vertically in a uniform magnetic field of strength 0.2 Weber/m^2. The coil carries a current of 2 A. If the plane of the coil is inclined at an angle of 30° with the direction of the field, the torque required to keep the coil in stable equilibrium will be :

(1) 0.24 Nm

(2) 0.12 Nm

(3) 0.15 Nm

(4) 0.20 Nm

14. An automobile moves on a road with a speed of 54 km h^{-1}. The radius of its wheels is 0.45 m and the moment of inertia of the wheel about its axis of rotation is 3 kg m^2. If the vehicle is brought to rest in 15 s, the magnitude of average torque transmitted by its brakes to the wheel is :

(1) 10.86 kg $m^2 s^{-2}$

2.86 kg $m^2 s^{-2}$

6.66 kg $m^2 s^{-2}$

8.58 kg $m^2 s^{-2}$

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15. Two metal wires of identical dimensions are connected in series. If σ_1 and σ_2 are the conductivities of the metal wires respectively, the effective conductivity of the combination is :

(1) $\frac{\sigma_1 + \sigma_2}{\sigma_1 \sigma_2}$

(3) $\frac{\sigma_1 + \sigma_2}{2\sigma_1 \sigma_2}$

(4) $\frac{2\sigma_1 \sigma_2}{\sigma_1 + \sigma_2}$

16. If potential (in volts) in a region is expressed as $V(x, y, z) = 6xy - y + 2yz$, the electric field (in N/C) at point $(1, 1, 0)$ is :

(1) $-\left(2\hat{i} + 3\hat{j} + \hat{k}\right)$

(2) $-\left(6\hat{i} + 9\hat{j} + \hat{k}\right)$

(3) $-\left(3\hat{i} + 5\hat{j} + 3\hat{k}\right)$

(4) $-\left(6\hat{i} + 5\hat{j} + 2\hat{k}\right)$

17. Two particles A and B, move with constant velocities $\vec{v}_1$ and $\vec{v}_2$. At the initial moment their position vectors are $\vec{r}_1$ and $\vec{r}_2$ respectively. The condition for particles A and B for their collision is :

(1) $\vec{r}_1 \times \vec{v}_1 = \vec{r}_2 \times \vec{v}_2$

(2) $\vec{r}_1 - \vec{r}_2 = \vec{v}_1 - \vec{v}_2$

(3) $\frac{\vec{r}_1 - \vec{r}_2}{|\vec{r}_1 - \vec{r}_2|} = \frac{\vec{v}_2 - \vec{v}_1}{|\vec{v}_2 - \vec{v}_1|}$

(4) $\vec{r}_1 \cdot \vec{v}_1 = \vec{r}_2 \cdot \vec{v}_2$

18. 4.0 g of a gas occupies 22.4 litres at NTP. The specific heat capacity of the gas at constant volume is 5.0 $JK^{-1} mol^{-1}$. If the speed of sound in this gas at NTP is 952 ms^{-1}, then the heat capacity at constant pressure is

(Take gas constant $R = 8.3\ JK^{-1} mol^{-1}$)

(1) 7.0 $JK^{-1} mol^{-1}$

(2) 8.5 $JK^{-1} mol^{-1}$

(3) 8.0 $JK^{-1} mol^{-1}$

(4) 7.5 $JK^{-1} mol^{-1}$

19. A force $\vec{F} = \alpha\hat{i} + 3\hat{j} + 6\hat{k}$ is acting at a point $\vec{r} = 2\hat{i} - 6\hat{j} - 12\hat{k}$. The value of α for which angular momentum about origin is conserved is :

(1) zero

(2) 1

(3) – 1

(4) 2

20. At the first minimum adjacent to the central maximum of a single-slit diffraction pattern, the phase difference between the Huygen's wavelet from the edge of the slit and the wavelet from the midpoint of the slit is :

(1) π radian

(2) $\frac{\pi}{8}$ radian

(3) $\frac{\pi}{4}$ radian

(4) $\frac{\pi}{2}$ radian

21. The heart of a man pumps 5 litres of blood through the arteries per minute at a pressure of 150 mm of mercury. If the density of mercury be 13.6×10^3 kg/m^3 and $g = 10$ m/s^2 then the power of heart in watt is :

(1) 3.0

(2) 1.50

(3) 1.70

(4) 2.35

22. A ball is thrown vertically downwards from a height of 20 m with an initial velocity v_0. It collides with the ground, loses 50 percent of its energy in collision and rebounds to the same height. The initial velocity v_0 is : (Take $g = 10\ ms^{-2}$)

(1) 28 ms^{-1}

(2) 10 ms^{-1}

(3) 14 ms^{-1}

(4) 20 ms^{-1}

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23. The cylindrical tube of a spray pump has radi R, one end of which has n fine holes, each of i r If the speed of the liquid in the tube is V, the speed of the ejection of the liquid through the holes i

(1) $\frac{VR^2}{n^3r^2}$

(2) $\frac{V^2R}{nr}$

(3) $\frac{VR^2}{nr}$

(4) $\frac{VR^2}{nr^2}$

24. A string is stretched between fixed points separated by 75.0 cm. It is observed to have resonant frequencies of 420 Hz and 315 Hz. There are no other resonant frequencies between these two. The lowest resonant frequency for this string is :

(1) 10.5 Hz

(2) 105 Hz

(3) 155 Hz

(4) 205 Hz

If dimensions of critical velocity v_c of a liquid flowing through a tube are expressed as $[\eta^x \rho^y r^z]$, where η, ρ and r are the coefficient of viscosity of liquid, density of liquid and radius of the tube respectively, then the values of x, y and z are given by :

(1) $-1, -1, -1$

(2) $1, 1, 1$

(3) $1, -1, -1$

(4) $-1, -1, 1$

26. A nucleus of uranium decays at rest into nuclei of

(1) thorium and helium. Th n :
than the thorium nucleus.

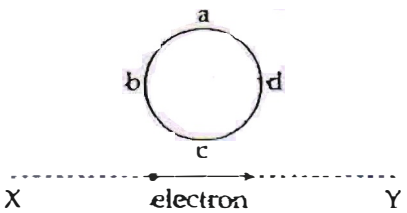
(2) The helium nucleus has less kinetic energy than the thorium nucleus.

(3) The helium nucleus has more kinetic energy than the thorium nucleus.

(4) The helium nucleus has less momentum the thorium nucleus

(1) The helium nucleus has more momentum

27. An electron moves on a straight line path XY as shown. The abcd is a coil adjacent to the path of electron. What will be the direction of current, if any, induced in the coil ?



(1) The current will reverse its direction as the electron goes past the coil

(2) No current induced

(3) abcd

(4) adcb

28. Water rises to a height 'h' in capillary tube. If the length of capillary tube above the surface of water is made less than 'h', then :

(1) water rises upto a point a little below the top and stays there.

(2) water does not rise at all.

(3) water rises upto the tip of capillary tube and then starts overflowing like a fountain.

water rises upto the top of capillary tube and stays there without overflowing.

29. In an astronomical telescope in normal adjustment a straight black line of length L is drawn on inside part of objective lens. The eye-piece forms a real image of this line. The length of this image is I. The magnification of the telescope is :

(1) $\frac{L+I}{L-I}$

(2) $\frac{L}{I}$

(3) $\frac{L}{I}+1$

(4) $\frac{L}{I}-1$

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30. A circuit contains an ammeter, a battery of 30 V and a resistance 40.8 ohm all connected in series. If the ammeter has a coil of resistance 480 ohm and a shunt of 20 ohm, the reading in the ammeter will be :

(1) 2 A

(2) 1 A

(3) 0.5 A

(4) 0.25 A

31. On a frictionless surface, a block of mass M moving at speed v collides elastically with another block of same mass M which is initially at rest. After collision the first block moves at an angle θ to its initial direction and has a speed $\frac{v}{3}$. The second block's speed after the collision is :

(1) $\frac{v}{\sqrt{2}}$

(2) $\frac{\sqrt{3}}{2}v$

(3) $\frac{2\sqrt{2}}{3}v$

(4) $\frac{3}{4}v$

32. A satellite S is moving in an elliptical orbit around the earth. The mass of the satellite is very small compared to the mass of the earth. Then,

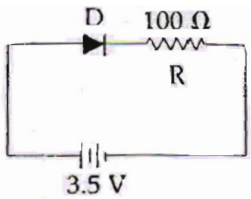
(1) the linear momentum of S remains constant in magnitude.

(2) the acceleration of S is always directed towards the centre of the earth.

(3) the angular momentum of S about the centre of the earth changes in direction, but its magnitude remains constant.

(4) the total mechanical energy of S varies periodically with time.

33. In the given figure, a diode D is connected to an external resistance R = 100 Ω and an e.m.f. of 3.5 V. If the barrier potential developed across the diode is 0.5 V, the current in the circuit will be :



(1) 20 mA

(2) 35 mA

(3) 30 mA

(4) 40 mA

34. A potentiometer wire of length L and a resistance r are connected in series with a battery of e.m.f. E_0 and a resistance r_1. An unknown e.m.f. E is balanced at a length l of the potentiometer wire. The e.m.f. E will be given by :

(1) $\frac{E_0\, l}{L}$

(2) $\frac{L\, E_0\, r}{(r + r_1)l}$

(3) $\frac{L\, E_0\, r}{l\, r_1}$

(4) $\frac{E_0\, r}{(r + r_1)\quad L}$

35. Two stones of masses m and 2 m are whirled in horizontal circles, the heavier one in a radius $\frac{r}{2}$ and the lighter one in radius r. The tangential speed of lighter stone is n times that of the value of heavier stone when they experience same centripetal forces. The value of n is :

(1) 4

(2) 1

(3) 2

(4) 3

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36. Two slits in Youngs experiment have widths in the ratio 1 : 25. The ratio of intensity at the maxima and minima in the interference pattern, $\frac{I_{max}}{I_{min}}$ is :

(1) $\frac{49}{121}$

(2) $\frac{4}{9}$

(3) $\frac{9}{4}$

(4) $\frac{121}{49}$

37. The Young's modulus of steel is twice that of brass. Two wires of same length and of same area of cross section, one of steel and another of brass are suspended from the same roof. If we want the lower ends of the wires to be at the same level, then the weights added to the steel and brass wires must be in the ratio of :

(1) 4 : 1

(2) 1 : 1

(3) 1 : 2

(4) 2 : 1

38. The coefficient of performance of a refrigerator is 5. If the temperature inside freezer is $-20°C$, the temperature of the surroundings to which it rejects heat is :

(1) 11°C

(2) 21°C

(3) 31°C

(4) 41°C

39. Light of wavelength 500 nm is incident on a metal with work function 2.28 eV. The de Broglie wavelength of the emitted electron is :

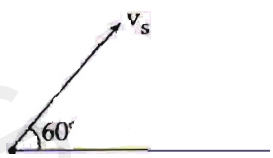
(1) $\geq 2.8 \times 10^{-9}$ m

(2) $\leq 2.8 \times 10^{-12}$ m

(3) $< 2.8 \times 10^{-10}$ m

(4) $< 2.8 \times 10^{-9}$ m

40. A source of sound S emitting waves of frequency 100 Hz and an observer O are located at some distance from each other. The source is moving with a speed of 19.4 ms^{-1} at an angle of 60° with the source observer line as shown in the figure. The observer is at rest. The apparent frequency observed by the observer (velocity of sound in air 330 ms^{-1}), is :



(1) 106 Hz

(2) 97 Hz

(3) 100 Hz

(4) 103 Hz

41. The value of coefficient of volume expansion of glycerin is $5 \times 10^{-4} K^{-1}$. The fractional change in the density of glycerin for a rise of 40°C in its temperature, is :

(1) 0.025

(2) 0.010

(3) 0.015

(4) 0.020

e position vector of a particle R as a function of ti e is given by :

$\vec{R} = 4 \sin(2\pi t)\ \hat{i} + 4 \cos(2\pi t)\ \hat{j}$

Where R is in meters, t is in seconds and $\hat{i}$ and $\hat{j}$ denote unit vectors along x-and y-directions, respectively. Which one of the following statements is **wrong** for the motion of particle ?

(1) Magnitude of the velocity of particle is 8 meter/second

(2) Path of the particle is a circle of radius 4 meter.

(3) Acceleration vector is along $-R$

(4) Magnitude of acceleration vector is $\frac{v^2}{R}$, where v is the velocity of particle.

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43. A plank with a box on it at one end is gradually raised about the other end. As the angle of inclination with the horizontal reaches 30°, the box starts to slip and slides 4.0 m down the plank in 4.0 s. The coefficients of static and kinetic friction between the box and the plank will be, respectively :



(1) 0.5 and 0.6

(2) 0.4 and 0.3

(3) 0.6 and 0.6

(4) 0.6 and 0.5

44. In the spectrum of hydrogen, the ratio of the longest wavelength in the Lyman series to the longest wavelength in the Balmer series is :

(1) $\frac{27}{5}$

(2) $\frac{5}{27}$

(3) $\frac{4}{9}$

$\frac{9}{4}$

45. A particle is executing a simple harmonic motion. Its maximum acceleration is α and maximum velocity is β. Then, its time period of vibration will be :

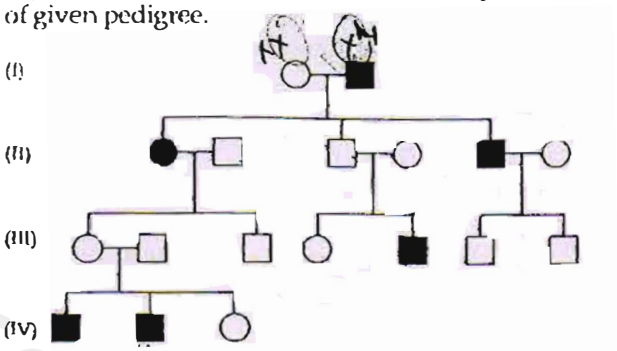
(1) $\frac{\beta^2}{\alpha}$

(2) $\frac{2\pi\beta}{\alpha}$

(3) $\frac{\beta^2}{\alpha^2}$

(4) $\frac{\alpha}{\beta}$

46. In the following human pedigree, the filled symbols represent the affected individuals. Identify the type of given pedigree.



(1) Autosomal recessive

(2) X-linked dominant

(3) Autosomal dominant

(4) X-linked recessive

47. Which of the following is not a function of the skeletal system ?

(1) Production of body heat

(2) Locomotion

(3) Production of erythrocytes

(4) Storage of minerals

Destruction of the anterior horn cells of the spinal cord would result in loss of :

(1) commissural impulses

(2) integrating impulses

(3) sensory impulses

(4) voluntary motor impulses

49. The term "linkage" was coined by

(1) G. Mendel

(2) W. Sutton

(3) T.H. Morgan

(4) T. Boveri

50. Filiform apparatus is characteristic feature of

(1) Aleurone cell

(2) Synergids

(3) Generative cell

(4) Nucellar embryo

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51. Satellite DNA is important because it :

(1) does not code for proteins and is same in all members of the population.

(2) codes for enzymes needed for DNA replication.

(3) codes for proteins needed in cell cycle.

(4) shows high degree of polymorphism in population and also the same degree of polymorphism in an individual, which is heritable from parents to children.

52. The wheat grain has an embryo with one large, shield-shaped cotyledon known as :

(1) Scutellum

(2) Coleoptile

(3) Epiblast

(4) Coleorrhiza

53. Identify the **correct** order of organisation of genetic material from largest to smallest :

(1) Genome, chromosome, gene, nucleotide

(2) Chromosome, genome, nucleotide, gene

(3) Chromosome, gene, genome, nucleotide

(4) Genome, chromosome, nucleotide, gene

54. Most animals that live in deep oceanic waters are :

(1) tertiary consumers

(2) detritivores

(3) primary consumers

(4) secondary consumers

55. Cell wall is absent in :

(1) Mycoplasma

(2) *Nostoc*

(3) *Aspergillus*

(4) *Funaria*

56. In which of the following interactions both partners are adversely affected ?

(1) Parasitism

(2) Mutualism

(3) Competition

(4) Predation

57. Human urine is usually acidic because :

(1) potassium and sodium exchange generates acidity.

(2) hydrogen ions are actively secreted into the filtrate.

(3) the sodium transporter exchanges one hydrogen ion for each sodium ion, in peritubular capillaries.

(4) excreted plasma proteins are acidic.

58. Match the following list of microbes and their importance :

(a)	*Sacharomyces cerevisiae*	(i)	Production of immunosuppressive agents
(b)	*Monascus purpureus*	(ii)	Ripening of Swiss cheese
(c)	*Trichoderma polysporum*	(iii)	Commercial production of ethanol
(d)	*Propionibacterium sharmanii*	(iv)	Production of blood-cholesterol lowering agents

	(a)	(b)	(c)	(d)
(1)	(iv)	(ii)	(i)	(iii)
(2)	(iii)	(i)	(iv)	(ii)
(3)	(iii)	(iv)	(i)	(ii)
(4)	(iv)	(iii)	(ii)	(i)

59. The body cells in cockroach discharge their nitrogenous waste in the haemolymph mainly in the form of :

(1) Urea

(2) Calcium carbonate

(3) Ammonia

(4) Potassium urate

60. In which of the following both pairs have **correct** combination ?

(1)	Gaseous nutrient cycle	Nitrogen and sulphur
	Sedimentary nutrient cycle	Carbon and Phosphorus
(2)	Gaseous nutrient cycle	Sulphur and Phosphorus
	Sedimentary nutrient cycle	Carbon and Nitrogen
(3)	Gaseous nutrient cycle	Carbon and Nitrogen
	Sedimentary nutrient cycle	Sulphur and Phosphorus
(4)	Gaseous nutrient cycle	Carbon and sulphur
	Sedimentary nutrient cycle	Nitrogen and Phosphorus

61. Which one is a wrong statement ?

(1) Haploid endosperm is typical feature of gymnosperms

(2) Brown algae have chlorophyll a and c, and fucoxanthin

(3) Archegonia are found in Bryophyta, Pteridophyta and Gymnosperms

(4) *Mucor* has biflagellate zoospores

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62. Match the columns and identify the correct option.

	Column I		Column II
(a)	Thylakoid		Disc-shaped sacs in Golgi apparatus
(b)	Cristae		Condensed structure of DNA
(c)	Cisterna		Flat membranous sacs in stroma
(d)	Chromatin	(iv)	Infoldings in mitochondria

	(a)	(b)	(c)	(d)
(1)	(iii)	(i)	(iv)	(ii)
(2)	(iii)	(iv)	(ii)	(i)
(3)	(iv)	(iii)	(i)	(ii)
(4)	(iii)	(iv)	(i)	(ii)

63. If you suspect major deficiency of antibodies in a person, to which of the following would you look for confirmatory evidence?
(1) Haemocytes
(2) Serum globulins
(3) Fibrinogin in plasma
(4) Serum albumins

64. The imperfect fungi which are decomposers of litter and help in mineral cycling belong to
(1) Phycomycetes
(2) Ascomycetes
(3) Deuteromycetes
(4) Basidiomycetes

65. The oxygen evolved during photosynthesis comes from water molecules. Which one of the following pairs of elements is involved in this reaction?
(1) Magnesium and Molybdenum
(2) Magnesium and Chlorine
(3) Manganese and Chlorine
(4) Manganese and Potassium

66. Body having meshwork of cells, internal cavities lined with food filtering flagellated cells and indirect development are the characteristics of phylum
(1) Mollusca
(2) Protozoa
(3) Coelenterata
(4) Porifera

67. Root pressure develops due to
(1) Passive absorption
(2) Increase in transpiration
(3) Active absorption
(4) Low osmotic potential in soil

68. The enzyme that is not present in succus enteric is :
(1) nucleosidase
(2) lipase
(3) maltase
(4) nucleases

69. Metagenesis refers to :
(1) Occurrence of a drastic change in form during post-embryonic development
(2) Presence of a segmented body an parthenogenetic mode of reproduction
(3) Presence of different morphic forms
(4) Alternation of generation between asexual and sexual phases of an organism

70. A protoplast is a cell :
(1) undergoing division
(2) without cell wall
(3) without plasma membrane
(4) without nucleus

71. The DNA molecule to which the gene of interest is integrated for cloning is called
(1) Template
(2) Carrier
(3) Transformer
(4) Vector

72. Which of the following structures is not found in a prokaryotic cell?
(1) Mesosome
(2) Plasma membrane
(3) Nuclear envelope
(4) Ribosome

73. The structures that help some bacteria to attach to rocks and/or host tissues are :
(1) Mesosomes
(2) Holdfast
(3) Rhizoids
(4) Fimbriae

74. Which one of the following hormones is not involved in sugar metabolism?
(1) Insulin
(2) Glucagon
(3) Cortisone
(4) Aldosterone

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75. In photosynthesis, the light-independent reactions take place at
(1) Photosystem II
(2) Stromal matrix
(3) Thylakoid lumen
(4) Photosystem I

76. The chitinous exoskeleton of arthropods is formed by the polymerisation of :
(1) N - acetyl glucosamine
(2) lipoglycans
(3) keratin sulphate and chond phate
(4) D -glucosamine

77. Select the wrong statement :
(1) The term *'contagium vivum fluidum'* was coined by M. W. Beijerinek
(2) Mosaic disease in tobacco and AIDS in human being are caused by viruses
(3) The viroids were discovered by D. J. Ivanowski
(4) W.M. Stanley showed that viruses could be crystallized

78. Among china rose, mustard, brinjal, potato, guava, cucumber, onion and tulip, how many plants have superior ovary ?
(1) Three
(2) Four
(3) Five
(4) Six

79. In angiosperms, microsporogenesis and megasporogenesis :
(1) Involve meiosis
(2) occur in ovule
(3) occur in anther
(4) form gametes without further div

(1) endoplasmic reticulum, ribosor
80. Cellular organelles with membranes are
(2) lysosomes, Golgi apparatus and mitochondria
(3) nuclei, ribosomes and mitochondria
(4) chromosomes, ribosomes and endoplasmic reticulum

81. Which one of the following is not applicable to RNA ?
(1) Heterocyclic nitrogenous bases
(2) Chargaff's rule
(3) Complementary base pairing
(4) 5' phosphoryl and 3' hydroxyl ends

82. Which of the following are most suitable indicators of SO_2 pollution in the environment ?
(1) Algae
(2) Fungi
(3) Lichens
(4) Conifers

83. Acid rain is caused by increase in the atmospheric concentration of :
(1) CO_2 and CO
(2) O_3 and dust
(3) SO_2 and NO_2
(4) SO_3 and CO

84. Which of the following immunoglobulins does constitute the largest percentage in human milk ?
(1) Ig A
(2) Ig G
(3) Ig D
(4) Ig M

85. Which of the following diseases is caused by a protozoan ?
(1) Babesiosis
(2) Blastomycosis
(3) Syphilis
(4) Influenza

86. Outbreeding is an important strategy of animal husbandry because it :
is useful in overcoming inbreeding depression.
(2) exposes harmful recessive genes that are eliminated by selection.
(3) helps in accumulation of superior genes.
(4) is useful in producing purelines of animals.

87. Which one of the following animals has two separate circulatory pathways ?
(1) Whale
(2) Shark
(3) Frog
(4) Lizard

88. Name the pulmonary disease in which alveolar surface area involved in gas exchange is drastically reduced due to damage in the alveolar walls.
(1) Pneumonia
(2) Asthma
(3) Pleurisy
(4) Emphysema

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89. A childless couple can be assisted to have a child through a technique called GIFT. The full form of this technique is
(1) Gamete internal fertilization and transfer
(2) Germ cell internal fallopian transfer
(3) Gamete inseminated fallopian transfer
(4) Gamete intra fallopian transfer

90. A gene showing codominance has :
(1) alleles that are recessive to each other
(2) both alleles independently expressed in the heterozygote
(3) one allele dominant on the other
(4) alleles tightly linked on the same chromosome

91. Doctors use stethoscope to hear the sounds produced during each cardiac cycle. The second sound is heard when :
Semilunar valves close down after the blood flows into vessels from ventricles
(2) AV node receives signal from SA node
(3) AV valves open up
(4) Ventricular walls vibrate due to gushing in of blood from atria

92. Read the different components from (a) to (d) in the list given below and tell the correct order of the components with reference to their arrangement from outer side to inner side in a woody dicot stem :
(a) Secondary cortex
(b) Wood
(c) Secondary phloem
(d) Phellem
The correct order is :
(1) (d), (a), (c), (b)
(2) (d), (c), (a), (b)
(3) (c), (d), (b), (a)
(4) (a), (b), (d), (c)

93. Which one of the following hormones though synthesised elsewhere, is stored and released by the master gland ?
(1) Prolactin
(2) Melanocyte stimulating hormone
(3) Antidiuretic hormone
(4) Luteinizing hormone

94. The wings of a bird and the wings of an insect are :
(1) phylogenetic structures and represent divergent evolution
homologous structures and represent convergent evolution
(3) homologous structures and represent divergent evolution
(4) analogous structures and represent convergent evolution

95. Which of the following pairs is not correctly matched ?

	Mode of reproduction	Example
(1)	Binary fission	*Sargassum*
(2)	Conidia	*Penicillium*
(3)	Offset	Water hyacinth
(4)	Rhizome	Banana

96. In his classic experiments on pea plants, Mendel did not use :
(1) Seed shape
(2) Flower position
(3) Seed colour
(4) Pod length

97. The function of the gap junction is to :
(1) separate two cells from each other
(2) stop substance from leaking across a tissue.
(3) performing cementing to keep neighbouring cells together.
facilitate communication between adjoining cells by connecting the cytoplasm for rapid large molecules.
transfer of ions, small molecules and some

98. Eutrophication of water bodies leading to killing fishes is mainly due to non-availability of :
(1) essential minerals
(2) oxygen
(3) food
(4) light

99. In human females, meiosis-II is not completed until ?
(1) uterine implantation
(2) birth
(3) puberty
(4) fertilization

100. The species confined to a particular region and not found elsewhere is termed as :
(1) Endemic
(2) Rare
(3) Keystone
(4) Alien

The UN conference of Parties on climate change in the year 2012 was held at :
(1) Lima
(2) Warsaw
(3) Durban
(4) Doha

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12. Which of the following layers in an antral follicle is acellular ?

(1) Stroma
(2) Zona pellucida
(3) Granulosa
(4) Theca interna

13. During biological nitrogen fixation, inactivation of nitrogenase by oxygen poisoning is prevented by :

(1) Carotene
(2) Cytochrome
(3) Leghaemoglobin
(4) Xanthophyll

14. Which of the following events is not associated with ovulation in human female ?

(1) Release of secondary oocyte
(2) LH surge
(3) Decrease in estradiol
(4) Full development of Graafian follicle

15. Arrange the following events of meiosis in correct sequence :

(a) Crossing over
(b) Synapsis
(c) Terminalisation of chiasmata
(d) Disappearance of nucleolus

(1) (a), (b), (c), (d)
(2) (b), (c), (d), (a)
(3) (b), (a), (d), (c)
(4) (b), (a), (c), (d)

16. Which of the following joints would allow no movement ?

(1) Synovial joint
(2) Ball and Socket joint
(3) Fibrous joint
(4) Cartilaginous joint

17. Flowers are unisexual in

(1) China rose
(2) Onion
(3) Pea
(4) Cucumber

108. A colour blind man marries a woman with normal sight who has no history of colour blindness in her family. What is the probability of their grandson being colour blind ?

(1) Nil
(2) 0.25
(3) 0.5
(4) 1

109. The cutting of DNA at specific locations became possible with the discovery of :

(1) Selectable markers
(2) Ligases
(3) Restriction enzymes
(4) Probes

110. Roots play insignificant role in absorption of water in :

(1) Pea
(2) Wheat
(3) Sunflower
(4) *Pistia*

111 Male gametophyte in angiosperms produces :

(1) Single sperm and two vegetative cells
(2) Three sperms
(3) Two sperms and a vegetative cell
(4) Single sperm and a vegetative cell

112. In which group of organisms the cell walls form two thin overlapping shells which fit together ?

(1) Dinoflagellates
(2) Slime moulds
(3) Chrysophytes
(4) Euglenoids

113. Coconut water from a tender coconut is :

(1) Innermost layers of the seed coat
(2) Degenerated nucellus
(3) Immature embryo
(4) Free nuclear endosperm

114. A column of water within xylem vessels of tall trees does not break under its weight because of :

(1) Lignification of xylem vessels
(2) Positive root pressure
(3) Dissolved sugars in water
(4) Tensile strength of water

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115 Chromatophores take part in :

(1) Movement

(2) Respiration

(3) Photosynthesis

(4) Growth

116. In mammalian eye, the 'fovea' is the center of the visual field, where :

(1) only rods are present.

(2) more rods than cones are found.

(3) high density of cones occur, but has no rods.

(4) the optic nerve leaves the eye.

117. Industrial melanism is an example of :

(1) Mutation

(2) Neo Lamarckism

(3) Neo Darwinism

(4) Natural selection

118. Which of the following biomolecules does have a phosphodiester bond ?

(1) Amino acids in a polypeptide

(2) Nucleic acids in a nucleotide

(3) Fatty acids in a diglyceride

(4) Monosaccharides in a polysaccharide

119. An association of individuals of different species living in the same habitat and having functional interactions is :

(1) Ecosystem

(2) Population

(3) Ecological niche

(4) Biotic community

120. Golden rice is a genetically modified crop plant where the incorporated gene is meant for biosynthesis of :

(1) Omega 3

(2) Vitamin A

(3) Vitamin B

(4) Vitamin C

121. The introduction of t-DNA into plants involves:

(1) Exposing the plants to cold for a brief peri

(2) Allowing the plant roots to stand in water

(3) Infection of the plant by *Agrobacterium tumefaciens*

(4) Altering the pH of the soil, then heat-shocking the plants

122. Auxin can be bioassayed by :

(1) Potometer

(2) Lettuce hypocotyl elongation

(3) Avena coleoptile curvature

(4) Hydroponics

123. Pick up the **wrong** statement :

(1) Some fungi are edible

(2) Nuclear membrane is present in Monera

(3) Cell wall is absent in Animalia

(4) Protista have photosynthetic and heterotrophic modes of nutrition

124. Ectopic pregnancies are referred to as :

(1) Implantation of defective embryo in the uterus

(2) Pregnancies terminated due to hormonal imbalance.

(3) Preg nancies with genetic abnormality.

(4) Implantation of embryo at site other than uterus.

125. Axile placentation is present in

(1) Pea

(2) *Argemone*

(3) *Dianthus*

(4) Lemon

126. The primary dentition in human differs from permanent dentition in not having one of the following type of teeth

(1) Molars

(2) Incisors

(3) Canine

(4) Premolars

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127. Which one of the following fruits is parthenocarpic?

(1) Jackfruit

(2) Banana

(3) Brinjal

(4) Apple

Balbiani rings are sites of :

(1) Polysaccharide synthesis

(2) RNA and protein synthesis

(3) Lipid synthesis

(4) Nucleotide synthesis

129. A pleiotropic gene :

controls a trait only in combination with another gene

controls multiple traits in an individual.

(3) is expressed only in primitive plants.

(4) is a gene evolved during Pliocene.

Grafted kidney may be rejected in a patient due to :

(1) Passive immune response

(2) Innate immune response

(3) Humoral immune response

(4) Cell-mediated immune response

131 During ecological succession :

(1) the numbers and types of animals remain constant.

(2) the changes lead to a community that is in near equilibrium with the environment and is called pioneer community

the gradual and predictable change in species composition occurs in a given area

(4) the establishment of a new biotic community is very fast in its primary phase.

132. Which of the following are **not** membrane - bound ?

(1) Lysosomes

(2) Mesosomes

(3) Vacuoles

(4) Ribosomes

133. Increase in concentration of the toxicant at successive trophic levels is known as :

(1) Biotransformation

(2) Biogeochemical cycling

(3) Biomagnification

(4) Biodeterioration

134. A jawless fish, which lays eggs in fresh water and whose ammocoetes larvae after metamorphosis return to the ocean is :

(1) *Neomyxine*

(2) *Petromyzon*

(3) *Eptatretus*

(4) *Myxine*

135. Choose the **wrong** statement :

(1) Morels and truffles are poisonous mushrooms

(2) Yeast is unicellular and useful in fermentation

(3) *Penicillium* is multicellular and produces antibiotics

(4) *Neurospora* is used in the study of biochemical genetics

136. Strong reducing behaviour of H_3PO_2 is due to

(1) High electron gain enthalpy of phosphorus

(2) High oxidation state of phosphorus

(3) Presence of two $-OH$ groups and one $P-H$ bond

(4) Presence of one $-OH$ group and two $P-H$ bonds

137. The stability of +1 oxidation state among Al, Ga, In and Tl increases in the sequence :

(1) Al < Ga < In < Tl

(2) Tl < In < Ga < Al

(3) In < Tl < Ga < Al

(4) Ga < In < Al < Tl

138. The oxidation of benzene by V_2O_5 in the presence of air produces :

(1) maleic anhydride

(2) benzoic acid

(3) benzaldehyde

(4) benzoic anhydride

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139. If the equilibrium constant for $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$ is K, the equilibrium constant for $\frac{1}{2}N_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons NO(g)$ will be :

(1) $\frac{1}{2}K$

(2) K

(3)

(4)

140. Gadolinium belongs to 4*f* series. It's atomic number is 64. Which of the following is the correct electronic configuration of gadolinium ?

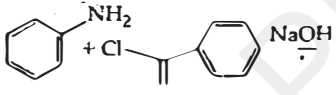
(1) [Xe] $4f^9 5s^1$

(2) [Xe] $4f^7 5d^1 6s^2$

(3) [Xe] $4f^6 5d^2 6s^2$

(4) [Xe] $4f^8 6d^2$

141. The following reaction



is known by the name

(1) Perkin's reaction

(2) Acetylation reaction

(3) Schotten-Baumen reaction

(4) Friedel-Craft's reaction

142. What is the mass of the precipitate formed when 50 mL of 16.9% so s mixed with 50 mL of 5.8% NaCl solution ?

(Ag = 107.8, N = 14, O = 16, Na = 23, Cl = 35.5)

(1) 3.5 g

(2)

(3) 14 g

(4) 28 g

143. What is the pH of the resulting solution when equal volumes of 0.1 M NaOH and 0.01 M HCl are mixed ?

(1) 2.0

(2) 7.0

(3) 1.04

(4) 12.65

144. Method by which Aniline cannot be prepared

(1) degradation of benzamide with bromine in alkaline solution.

(2) reduction of nitrobenzene with H_2/Pd in ethanol.

(3 potassium salt of phthalimide treated wi chlorobenzene followed by hydrolysis wi[A] aqueous NaOH solution.

(4) hydrolysis of phenylisocyanide with acid solution

145. If Avogadro number N_A, is changed fro 6.022×10^{23} mol^{-1} to 6.022×10^{20} mol^{-1}, this would ch

(1 the mass of one mole of carbon.

(2) the ratio of chemical species to each other in a balanced equation.

(3) the ratio of elements to each other in a compound.

(4) the definition of mass in units of grams.

146. The variation of the boiling points of the hydrogen halides is in the order HF > HI > HBr > HCl.

What explains the higher boiling point of hydrogen fluoride ?

(1) There is strong hydrogen bonding between HF molecules.

(2) The bond energy of HF molecules is greater than in other hydrogen halides.

(3) The effect of nuclear shielding is much reduced in fluorine which polarises the HF molecule.

(4) The electronegativity of fluorine is much higher than for other elements in the group.

147. The number of structural isomers possible from the molecular formula C_3H_9N is :

(1) 5

(2) 2

(3) 3

(4) 4

148. Which of the statements given below is incorrect ?

(1) O_3 molecule is bent

(2) ONF is isoelectronic with O_2N^-

(3) OF_2 is an oxide of fluorine

(4) Cl_2O_7 is an anhydride of perchloric acid

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O

The formation of the oxide ion, $O^{2-}(g)$, from oxygen atom requires first an exothermic and then an endothermic step as shown below :

$O(g) + e^- \rightarrow O^-(g)$; $\Delta_f H^\ominus = -141$ kJ mol^{-1}

$O^-(g) + e^- \rightarrow O^{2-}(g)$; $\Delta_f H^\ominus = +780$ kJ mol^{-1}

Thus process of formation of O^{2-} in gas phase is unfavourable even though O^{2-} is isoelectronic with neon. It is due to the fact that,

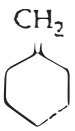
(1) O^- ion has comparatively smaller size than oxygen atom.

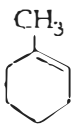
(2) oxygen is more electronegative

(3) addition of electron in oxygen results in larger size of the ion.

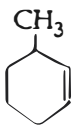
(4) electron repulsion outweighs the stability gained by achieving noble gas configuration.

1. In the reaction with HCl, an alkene reacts in accordance with the Markovnikov's rule, to give a product 1-chloro-1-methylcyclohexane. The possible alkene is

(1)  (A)

(2)  (B)

(3) (A) and (B)

(4) 

1. 2,3 - Dimethyl-2-butene can be prepared by heating which of the following compounds with a strong acid ?

(1) $(CH_3)_3C-CH=CH_2$

(2) $(CH_3)_2C=CH-CH_2-CH_3$

(3) $(CH_3)_2CH-CH_2-CH=CH_2$

(4) $(CH_3)_2CH-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH=CH_2$

2. The hybridization involved in complex $[Ni(CN)_4]^{2-}$ is : (At. No. Ni = 28)

(1) sp^3

(2) d^2sp^2

(3) d^2sp^3

(4) dsp^2

153. Reaction of a carbonyl compound with one of the following reagents involves nucleophilic addition followed by elimination of water. The reagent is :

(1) hydrazine in presence of feebly acidic solution

(2) hydrocyanic acid

(3) sodium hydrogen sulphite

(4) a Grignard reagent

154. Two possible stereo-structures of $CH_3CHOH.COOH$, which are optically active, are called :

(1) Atropisomers

(2) Enantiomers

(3) Mesomers

(4) Diastereomers

155. In the extraction of copper from its sulphide ore, the metal is finally obtained by the reduction of cuprous oxide with :

(1) carbon monoxide

(2) copper(I) sulphide

(3) sulphur dioxide

(4) iron(II) sulphide

156. A gas such as carbon monoxide would be most likely to obey the ideal gas law at :

(1) low temperatures and high pressures.

(2) high temperatures and high pressures.

(3) low temperatures and low pressures.

(4) high temperatures and low pressures.

157. In an S_N1 reaction on chiral centres, there is :

(1) inversion more than retention leading to partial racemization

(2) 100% retention

(3) 100% inversion

(4) 100% racemization

158. The vacant space in bcc lattice unit cell is :

(1) 48%

(2) 23%

(3) 32%

(4) 26%

159. The name of complex ion, $[Fe(CN)_6]^{3-}$ is :

(1) Hexacyanitoferrate (III) ion

(2) Tricyanoferrate (III) ion

(3) Hexacyanidoferrate (III) ion

(4) Hexacyanoiron (III) ion

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160. The number of water molecules is maximum in :

(1) 1.8 gram of water
(2) 18 gram of water
(3) 18 moles of water
(4) 18 molecules of water

161. The heat of combustion of carbon to CO_2 is -393.5 kJ/mol. The heat released upon formation of 35.2 g of CO_2 from carbon and oxygen gas is :

(1) +315 kJ
(2) −630 kJ
(3) −3.15 kJ
(4) −315 kJ

162. Aqueous solution of which of the following compounds is the best conductor of electric current ?

(1) Hydrochloric acid, HCl
(2) Ammonia, NH_3
(3) Fructose, $C_6H_{12}O_6$
(4) Acetic acid, $C_2H_4O_2$

163. In which of the following pairs, both the species are not isostructural ?

(1) diamond, silicon carbide
(2) NH_3, PH_3
(3) XeF_4, XeO_4
(4) $SiCl_4$, PCl_4^+

164. What is the mole fraction of the solute in a 1.00 m aqueous solution ?

(1) 1.770
(2) 0.0354
(3) 0.0177
(4) 0.177

165. Which of the following statements is not correct for a nucleophile ?

(1) Ammonia is a nucleophile
(2) Nucleophiles attack low e^- density sites
(3) Nucleophiles are not electron seeking
(4) Nucleophile is a Lewis acid

166. The sum of coordination number and oxidation number of the metal M in the complex $[M(en)_2(C_2O_4)]Cl$ (where *en* is ethylenediamine) is :

(1) 6
(2) 7
(3) 8
(4) 9

167. Assuming complete ionization, same mol which of the following compounds will requir *least* amount of acidified $KMnO_4$ for compl oxidation ?

(1) $FeSO_3$
(2) FeC_2O_4
$Fe(NO_2)_2$
(4) $FeSO_4$

168. Reaction of phenol with chloroform in presence dilute sodium hydroxide finally introduces whi one of the following functional group ?

(1) $-COOH$
(2) $-CHCl_2$
(2) $-CHO$
(3) $-CH_2Cl$

169. Which is the correct order of increasing energy of the listed orbitals in the atom of titanium ?
(At. no. Z = 22)

(1) 4s 3s 3p 3d
(2) 3s 3p 3d 4s
(3) 3s 3p 4s 3d
(4) 3s 4s 3p 3d

170. Which one of the following pairs of solution is not an acidic buffer ?

(1) CH_3COOH and CH_3COONa
(2) H_2CO_3 and Na_2CO_3
(3)
(4) $HClO_4$ and $NaClO_4$

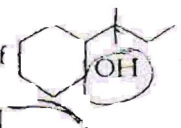
171. Number of possible isomers for the complex $[Co(en)_2Cl_2]Cl$ will be : (en = ethylenediamine)

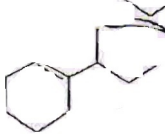
(1) 1
(2) 3
(3) 4
(4) 2

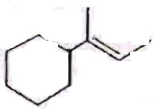
172. Decreasing order of stability of O_2, O_2^-, O_2^+ and O_2^{2-} is :

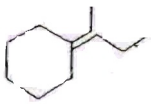
(1) $O_2^{2-} > O_2^- > O_2 > O_2^+$
(2) $O_2 > O_2^+ > O_2^{2-} > O_2^-$
(3) $O_2^- > O_2^{2-} > O_2^+ > O_2$
(4) $O_2^+ > O_2 > O_2^- > O_2^{2-}$

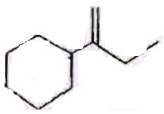
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173. Which of the following is not the product of dehydration of  ?

(1) 

(2) 

(3) 

(4) 

174. The correct statement regarding defects in crystalline solids is :

(1) Frenkel defects decrease the density of crystalline solids.

(2) Frenkel defect is a dislocation defect.

(3) Frenkel defect is found in halides of alkaline metals.

(4) Schottky defects have no effect on the of crystalline solids.

175. The rate constant of the reaction A → B is mole per second. If the concentratio then concentration of B after 20 minutes

(1) 3.60 M

(2) 0.36 M

(3) 0.72 M

(4) 1.08 M

176. On heating which of the following releas most easily ?

(1) Na_2CO_3

(2) $MgCO_3$

(3) $CaCO_3$

(4) K_2CO_3

177. 20.0 g of a magnesium carbona e sample d on heating to give carbon dioxi magnesium oxide. What will be purity of magnesium carbonate i

(1) 96

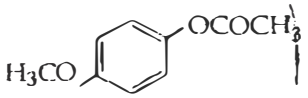
(2) 60

(3) 84

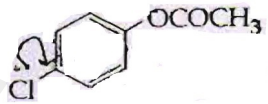
(4) 75

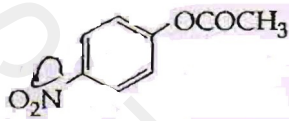
(At. Wt. : Mg = 24)

178. Which one of the following esters gets hydrolysed *most easily* under alkaline conditions ?

(1) 

(2) 

(3) 

(4) 

179. Caprolactam is used for the manufacture of :

(1) Teflon

(2) Terylene

(3) Nylon - 6, 6

(4) Nylon - 6

180. Which of the following reaction(s) can be used for the preparation of alkyl halides ?

(I) $CH_3CH_2OH + HCl \xrightarrow{\text{anh. } ZnCl_2}$

(II) $CH_3CH_2OH + HCl \longrightarrow$

(III) $(CH_3)_3COH + HCl \longrightarrow$

(IV) $(CH_3)_2CHOH + HCl$

(1) (I) and (II) only

(2) (IV) only

(3) (III) and (IV) only

(4) (I), (III) and (IV) only

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